

Multivariable

(Assignment 3)

Q) Laplace Transform (6.1)

Example 1.

Let $f(t) = 1$ when $t \geq 0$. Find $F(s)$.

$$f(t) = L^{-1}(F).$$

$$F(s) = L(f).$$

$$F(s) = L(f) = \int_0^{\infty} k(s,t) f(t) dt.$$

$$* k(s,t) = e^{-st}.$$

So

$$L(f) = L(1) = \int_0^{\infty} e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_0^{\infty} = \frac{1}{s}$$

$$\int_0^{\infty} e^{-st} f(t) dt = \lim_{T \rightarrow \infty} \int_0^T e^{-st} f(t) dt.$$

Hence

$$\int_0^{\infty} e^{-st} dt = \lim_{T \rightarrow \infty} \left[-\frac{1}{s} e^{-st} \right]_0^T.$$

$$= \lim_{T \rightarrow \infty} \left[-\frac{1}{s} e^{-sT} + \frac{1}{s} e^0 \right] = \frac{1}{s}$$

Example 2

Laplace Transform $\mathcal{L}(e^{at})$ of the exponential function e^{at} .

Let $f(t) = e^{at}$ when $t \geq 0$, where a is a constant. find $\mathcal{L}(f)$

Sol

$$\mathcal{L}(e^{at}) = \int_0^{\infty} e^{-st} e^{at} dt = \frac{1}{a-s} e^{-(s-a)t} \Big|_0^{\infty};$$

$$\mathcal{L}(e^{at}) = \frac{1}{s-a}$$

Ex 6.3.

Q) t ($0 < t < 2$).

In term of unit step function

$$f(t) = t(u(t) - u(t-2))$$

Transform of the function

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \mathcal{L}\{t(u(t)) - t(u(t-2))\} \\ &= \mathcal{L}\{t u(t)\} - \mathcal{L}\{t u(t-2)\} \end{aligned}$$

$$= \frac{1}{s^2} - e^{-2s} \left(\frac{1}{s^2} - \frac{2}{s} \right)$$

$$= \frac{1}{s^2} - \frac{e^{-2s}}{s^2} + \frac{2e^{-2s}}{s}$$

$$= \frac{1 - e^{-2s} + 2se^{-2s}}{s^2}$$

Q) $t - 2 (t > 2).$

$$f(t) = (t-2) u(t-2).$$

Transform

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{(t-2) u(t-2)\}.$$

$$\mathcal{L}\{f(t-a) u(t-a)\} = e^{-as} \mathcal{L}\{f(t)\}$$

so

$$= \frac{e^{-2s}}{s^2}.$$

Q) $\cos 4t (0 < t < \pi).$

$$f(t) = \cos 4t (u(t) - u(t-\pi)).$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\cos 4t u(t)\} - \mathcal{L}\{\cos 4t u(t-\pi)\}.$$

$$= \frac{s}{s^2-4} - e^{-\pi s} \left(\frac{s}{s^2-4} \right).$$

$$= \frac{s}{s^2-4} (1 - e^{-\pi s}).$$

Questions of Fourier Series.

$$f(x) = \begin{cases} 0, & -\pi < x < -\pi/2 \\ 1, & -\pi/2 < x < \pi/2 \\ 0, & \pi/2 < x < \pi. \end{cases}$$

let,

$$f(x) = \frac{1}{2} \cdot a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx).$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \sin nx dx.$$

for a_0 ,

$$a_0 = \frac{1}{\pi} \left\{ \int_{-\pi}^{-\pi/2} 0 dx + \int_{-\pi/2}^{\pi/2} 1 dx + \int_{\pi/2}^{\pi} 0 dx \right\}$$

$$= \frac{1}{\pi} \left(0 + \int_{-\pi/2}^{\pi/2} 1 dx + 0 \right)$$

$$= \frac{1}{\pi} \left[\frac{x}{1} + \frac{x}{1} \right]_{-\pi/2}^{\pi/2} = \frac{2\pi}{2\pi} = 1.$$

for a_n ,

$$a_n = \frac{1}{\pi} \left\{ 0 + \int_{-\pi/2}^{\pi/2} \cos nx dx + 0 \right\}$$

$$= \frac{1}{\pi} \left[\frac{\sin nx}{n} \right]_{-\pi/2}^{\pi/2} \Rightarrow \frac{1}{n\pi} \left\{ \sin \frac{n\pi}{2} - \sin \left(-\frac{n\pi}{2} \right) \right\}$$

$$= \frac{1}{n\pi} \cdot 2 \sin n \frac{\pi}{2} = \frac{2}{n\pi}$$

for b_n ,

$$b_n = \frac{1}{\pi} \left[\int_0^{\pi/2} 0 + \int_{-\pi/2}^{\pi/2} \sin n x dx + 0 \right]$$

$$= \frac{1}{n\pi} \left[-\cos n x \right]_{-\pi/2}^{\pi/2}$$

$$= -\frac{1}{n\pi} \left[\cancel{\cos n \frac{\pi}{2}} - \cancel{\cos n \frac{\pi}{2}} \right]$$

$$= -\frac{1}{n\pi} = 0.$$

Now,

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$= \frac{1}{2} + \frac{2}{\pi} \cos x + \frac{\cos 2x}{\pi} + \frac{2}{3\pi} \cos 3x + \dots$$

Question.

$$F(x) = \begin{cases} 0 & \text{if } -2 < x < -1 \\ K & \text{if } -1 < x < 1 \\ 0 & \text{if } 1 < x < 2 \end{cases} \quad p = 2L = 4, L =$$

Solution.

$$F(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$a_0 = \frac{1}{2L} \int_{-L}^L F(x) dx$$

$$a_n = \frac{1}{L} \int_{-L}^L F(x) \cos \frac{n\pi x}{L} dx$$

$$b_n = \frac{1}{L} \int_{-L}^L F(x) \sin \frac{n\pi x}{L} dx.$$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx.$$

$$a_0 = \frac{1}{2(2)} \left(\int_{-2}^{-1} 0 dx + \int_{-1}^1 K dx + \int_1^2 0 dx \right)$$

$$a_0 = \frac{1}{4} \left(\int_{-1}^1 K dx \right)$$

$$a_0 = \frac{K}{4} \int_{-1}^1 dx$$

$$a_0 = \frac{K}{4} \left[x \right]_{-1}^1$$

$$a_0 = \frac{K}{4} (1 - (-1))$$

$$a_0 = \frac{K}{4} (2) \Rightarrow a_0 = \frac{K}{2}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx$$

$$a_n = \frac{1}{2} \left(\int_{-1}^1 K \cos \frac{n\pi x}{2} dx \right)$$

$$a_n = \frac{K}{2} \left(\int_{-1}^1 \cos \frac{n\pi x}{2} dx \right)$$

$$a_n = \frac{K}{2} \left[\frac{\sin(n\pi x/2)}{n\pi/2} \right]_{-1}^1$$

$$a_n = \frac{K}{2} \left[\sin\left(\frac{n\pi}{2}\right) - \sin\left(-\frac{n\pi}{2}\right) \right] \frac{2}{n\pi}$$

$$a_n = \frac{K}{n\pi} 2 \sin\left(\frac{n\pi}{2}\right) \Rightarrow a_n = \frac{2K}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

putting values of n in a_n .

$$n=1$$

$$a_1 = \frac{2K}{\pi} \sin\left(\frac{\pi}{2}\right) \Rightarrow a_1 = \frac{2K}{\pi}$$

$$n=2$$

$$a_2 = \frac{2K}{2\pi} \sin\left(\frac{2\pi}{2}\right) \Rightarrow a_2 = \frac{K}{\pi} (0) \Rightarrow a_2 = 0$$

$$n=3$$

$$a_3 = \frac{2K}{3\pi} \sin\left(\frac{3\pi}{2}\right) \Rightarrow a_3 = \frac{2K}{3\pi} (-1) \Rightarrow a_3 = -\frac{2K}{3\pi}$$

$$n=4$$

$$a_4 = \frac{2K}{4\pi} \sin\left(\frac{4\pi}{2}\right) \Rightarrow a_4 = \frac{K}{2\pi} (0) \Rightarrow a_4 = 0$$

$$n=5$$

$$a_5 = \frac{2K}{5\pi} \sin\left(\frac{5\pi}{2}\right) \Rightarrow a_5 = \frac{2K}{5\pi} (1) \Rightarrow a_5 = \frac{2K}{5\pi}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx.$$

$$b_n = \frac{1}{2} \int_{-1}^1 K \sin \frac{n\pi x}{2} dx$$

$$b_n = \frac{K}{2} \frac{2}{n\pi} \left[-\cos \frac{n\pi x}{2} \right]_{-1}^1$$

$$b_n = -\frac{K}{n\pi} \left[+\cos \frac{n\pi}{2} - \cos \left(-\frac{n\pi}{2} \right) \right]$$

Now, put the values of n in b_n .

$$n=1$$

$$b_1 = \frac{-K}{\pi} \left[\cos \frac{\pi}{2} - \cos \left(-\frac{\pi}{2} \right) \right] = 0$$

$$n=2$$

$$b_2 = \frac{-K}{2\pi} \left[\cos \left(\frac{2\pi}{2} \right) - \cos \left(-\frac{2\pi}{2} \right) \right] = 0$$

$$n=3$$

$$b_3 = \frac{-K}{3\pi} \left[\cos \left(\frac{3\pi}{2} \right) - \cos \left(-\frac{3\pi}{2} \right) \right] = 0$$

$$n=4$$

$$b_4 = \frac{-K}{4\pi} \left[\cos \left(\frac{4\pi}{2} \right) - \cos \left(-\frac{4\pi}{2} \right) \right] = 0$$

$$n=5$$

$$b_5 = \frac{-K}{5\pi} \left[\cos \left(\frac{5\pi}{2} \right) - \cos \left(-\frac{5\pi}{2} \right) \right] = 0$$

Now, putting all the values in $F(x)$,

$$F(x) = a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L}}{L} \right)$$

$$F(x) = \frac{K}{2} + \frac{2K}{\pi} \cos \frac{\pi x}{2} + \left(\frac{-2K}{3\pi} \right) \cos \frac{3\pi x}{2} + \frac{2K}{5\pi} \cos \frac{5\pi x}{2} + \dots$$

$$F(x) = \frac{K}{2} + \frac{2K}{\pi} \left[\cos \left(\frac{\pi x}{2} \right) - \frac{1}{3} \cos \left(\frac{3\pi x}{2} \right) + \frac{1}{5} \cos \left(\frac{5\pi x}{2} \right) + \dots \right]$$